CARL WENDY SIMEA

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Computer Engineering Technology student who builds Arduino, Raspberry Pi and AI projects, from soldering circuits to writing C/C++ and Python code. Nearly two years of tech support and sales at T-Mobile; speaks English, Haitian Creole, French and Spanish. Seeking internships in embedded systems, robotics and hardware engineering.

EDUCATION

New York City College of Technology, CUNY — Brooklyn, NY
Bachelor of Technology, Computer Engineering Technology — Expected 12/2027

Relevant Coursework:

Logic and Problem Solving	Fundamentals of Digital Systems	Operating Systems Laboratory
Electronics	Software for Computer Control	Computer Hardware Systems
Electrical Circuits	Data Communications	Electromechanical Manufacturing Lab

Online Coursework: Harvard CS50x and CS50P (completed) · MIT 6.036 Introduction to Machine Learning (in progress)

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, JavaScript, SQL, HTML/CSS, x86 Assembly
Embedded Systems: Arduino (Uno, Mega 2560), Raspberry Pi 5, Servo and Motor Control, Rotary Encoders, PCB Design
Electronics: Circuit Assembly, Breadboarding, Soldering, Oscilloscope, Function Generator, Multimeter, DC Power Supply
Hardware & IT: PC Assembly, Hardware Troubleshooting, OS Installation, iOS and Android Troubleshooting, Data Migration
Fabrication: 3D Printing (Bambu Lab), Drill Press, Mechanical Measuring Tools
Software & Tools: Flask, SQLite, Ollama, Arduino IDE, Visual Studio, VS Code, GitHub, NI Multisim, Eclipse, Cygwin
Office: Microsoft Word, Excel, Outlook, Google Docs
Operating Systems: Windows, Linux (Ubuntu, Raspberry Pi OS), macOS, iOS, Android
Languages: English, Haitian Creole, French, Spanish

ACADEMIC PROJECTS

New York City College of Technology, Department of Computer Engineering Technology — Brooklyn, NY

Claw-Style Vending Machine (Team Project) — 09/2025 – 12/2025
- Built a claw-style vending machine with a team, letting the user steer the claw with buttons to grab and dispense an item.
- Ran the machine on an Arduino microcontroller that reads the button inputs, drives the motors and updates the display.

Computer Assembly and Integration — 01/2025 – 05/2025
- Assembled and configured computer systems, installing CPUs, RAM, motherboards and storage drives.
- Diagnosed and resolved hardware compatibility issues during assembly and integration.

Electromechanical Manufacturing Lab — 01/2024 – 05/2024
- Designed and built a digital trainer for testing and configuring logic gates.
- Shaped, fabricated and soldered the trainer's parts, then tested and troubleshot it until it worked properly.

Mechanisms Lab — 01/2024 – 05/2024
- Reverse engineered and repaired a Radio Shack Armatron robotic arm, fully disassembling and reassembling it.
- Worked with teammates to troubleshoot mechanical issues and implement fixes.

PERSONAL & HACKATHON PROJECTS

MedPeek: AI X-Ray Reader (Hacktoberfest Hack Day Brooklyn) — 10/2026
- Co-built a web app with two teammates where Google's Gemma 4 explains X-ray, MRI and CT images in plain language.
- Built a solo offline version in Flask and SQLite that runs Gemma 4 through Ollama, with user accounts and chat history.
- Designed it to compare uploads with normal FracAtlas hand X-rays, naming the broken bone, break type and likely cause.

Raspberry Pi Handheld Camera — In progress
- Building the controls around a single rotary encoder: press, double press, long press and scroll.
- Designed a four-state control flow (off, idle, timer setup, recording), with video saved to a 5-minute buffer.

Circuit Solver Web App (Harvard CS50x Final Project)
- Built a web app that calculates resistance in DC circuits, with a Python (Flask) back end and a JavaScript front end.

EXPERIENCE

T-Mobile USA, Inc. — New York, NY
Mobile Expert — 09/2024 – 07/2026
- Troubleshot Android and iOS devices for customers, resolving device and operating system issues.
- Sold phones and new lines with Retail Mobility tools, matching customers to the devices and plans that fit their needs.
- Performed data migrations between devices so customers kept all of their data when upgrading.